

AI Swarm Drones Find Survivors 3-4x Faster Than Traditional SAR

10-18 MIN/MI²

78-85% DETECTION

12% FALSE ALARMS

₹95,000 SORTIE



50Hz OFFBOARD

SUBSYSTEM A // GNC

SUTRA-FSD GNC & 3D ORCA

SOLVES PROBLEM VOID:
Canopy GPS Multi-Path Drift & Crashes

SOLUTIONAL MOAT:
Decentralized 3D velocity obstacle collision avoidance + PX4 MicroXRCE 50Hz offboard setpoint streaming.

0.05s
Response Latency

0 Tower
Dependency

DECENTRALIZED VIO50Hz RATE



-5dB RESILIENT

SUBSYSTEM B // COMMS

Deep JSCC Neural Video

SOLVES PROBLEM VOID:
Mountain Ravine RF Ridge Blackout

SOLUTIONAL MOAT:
Differentiable joint source-channel coding yielding zero digital cliff and graceful video degradation down to -5dB.

-5dB
SNR Threshold

0 Cliff
Degradation

SEMANTIC COMMSEDGE TPU



3.59cm ACCURACY

SUBSYSTEM C // VISION

Tri-Modal AI & 3D Geolocation

SOLVES PROBLEM VOID:
Smoke Blindspots & 15-30m Slope Drift

SOLUTIONAL MOAT:
Cross-attention fusion (RGB + FLIR Thermal + Radar) + 3D DEM elevation raycasting on Jetson Orin for 3.59cm GPS accuracy.

<15ms
Inference Time

3.59cm
Geo Error

TRI-MODAL FUSEDJETSON ORIN



1-2 OPERATORS

SUBSYSTEM D // C2 GCS

Pegasus 3D WebGPU GCS

SOLVES PROBLEM VOID:
15-25 Ground Crew Cognitive Overload

SOLUTIONAL MOAT:
Offline tactical 3D digital twin + ATAK plugin, empowering 1-2 operators to orchestrate full drone swarms.

1-2 Ops
Crew Size

100%
Offline ATAK

ATAK INTEGRATIONWEBGPU 3D